AC2204 Marking Scheme

Introduction to Management Accounting · Autumn 2021 · Paper 1

Examination Session and Year	Autumn 2021
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Ivo De Loo
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Separate answer books are not required.

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Fellows Ltd	Traditional (absorption) costing vs activity-based costing (batch-driven set-ups), relevant range and long-term decision-making	40
Q2 (Section B option)	Contessa Ltd	Marginal vs absorption costing, profit reconciliation, planning vs control	30
Q3 (Section B option)	Quaver Ltd	CVP analysis, operating leverage, incremental analysis (apprentice hire), multi-product WACM (new guitar line)	30
Q4 (Section B option)	Statement evaluation	Conversion costs, sunk vs opportunity vs differential costs, operating leverage vs margin of safety, high-low method, cost structure vs sales mix	30

QUESTION 1

Fellows Ltd

Total: 40 marks

Fellows Ltd produces three products and currently absorbs production overheads using a machine hour rate. Candidates are required to (A) recalculate product costs under both traditional and activity-based costing — where, unusually, the number of production batches (set-ups) must first be derived from total production volume and batch size — and (B) evaluate a management view about the relevance of the relevant range concept for long-term decision-making.

Part	Requirement	Marks
(A)	Total costs and cost per unit – traditional costing and activity-based costing	32
(B)	Agree/disagree discussion: does the relevant range matter given a long-term strategic view?	8

Question 1(A) **32 marks**

Fellows Ltd, most recent financial period. Product A: 600 units, €25 DM/unit, €100 DL/unit, 1 machine hour/unit, batch size 200, 6 purchase orders. Product B: 900 units, €32 DM/unit, €25 DL/unit, 3 machine hours/unit, batch size 100, 9 purchase orders. Product C: 300 units, €17 DM/unit, €50 DL/unit, 2 machine hours/unit, batch size 50, 10 purchase orders. Cost pools: Machine maintenance €35,100 (machine hours); Production scheduling €29,250 (production batches/set-ups); Quality €35,100 (units of output); Materials handling €17,550 (purchase orders). (A) Calculate the total costs and cost per unit for each product using (i) Traditional (absorption) costing and (ii) Activity-based costing (ABC).

Full Model Answer – (i) Traditional (absorption) costing

Direct labour is given directly as a cost per unit for each product (not derived from hours × rate), so no separate labour rate calculation is required. Total production overheads = €35,100 + €29,250 + €35,100 + €17,550 = €117,000. Total machine hours = (600 × 1) + (900 × 3) + (300 × 2) = 600 + 2,700 + 600 = 3,900. Overhead absorption rate = €117,000 ÷ 3,900 = €30 per machine hour.

Traditional (absorption) costing – cost per unit

Item	Product A	Product B	Product C
Direct materials	25.00	32.00	17.00
Direct labour	100.00	25.00	50.00
Overhead (MH/unit × €30)	30.00	90.00	60.00
Total cost per unit	**155.00**	**147.00**	**127.00**

Traditional (absorption) costing – total costs

Item	Product A (€)	Product B (€)	Product C (€)
Units produced	600	900	300
Total cost (unit cost × units)	**93,000**	**132,300**	**38,100**

Check: Total overhead absorbed = €18,000 (A) + €81,000 (B) + €18,000 (C) = €117,000, agreeing with the total production overhead given.

Full Model Answer – (ii) Activity-based costing

Unlike the previous questions in this series, the number of production batches (set-ups) per product is not given directly and must first be derived from total production volume divided by batch size.

Working – number of production batches (set-ups)

Product	Total production	Batch size	Number of batches
A	600	200	3
B	900	100	9
C	300	50	6
Total			**18**

Working – cost pools and activity rates

Cost pool	Pool amount (€)	Cost driver	Total activity	Rate
Machine maintenance	35,100	Machine hours	3,900 MH	€9.00 per MH
Production scheduling	29,250	Production batches	18 batches	€1,625.00 per batch
Quality	35,100	Units of output	1,800 units	€19.50 per unit
Materials handling	17,550	Purchase orders	25 orders	€702.00 per order
Total	**117,000**			

Full Model Answer – (ii) Activity-based costing (continued)

Working – overhead allocated to each product

Cost pool	Product A (€)	Product B (€)	Product C (€)
Machine maintenance (total MH × €9.00)	5,400	24,300	5,400
Production scheduling (batches × €1,625)	4,875	14,625	9,750
Quality (units × €19.50)	11,700	17,550	5,850
Materials handling (orders × €702)	4,212	6,318	7,020
Total overhead	**26,187**	**62,793**	**28,020**

Check: Total overhead allocated = €26,187 + €62,793 + €28,020 = €117,000, agreeing with the total production overhead given.

Activity-based costing – total costs and cost per unit

Item	Product A	Product B	Product C
Direct materials (total)	15,000	28,800	5,100
Direct labour (total)	60,000	22,500	15,000
Overhead (from above)	26,187	62,793	28,020
Total cost	**101,187**	**114,093**	**48,120**
Units produced	600	900	300
Cost per unit	**168.65**	**126.77**	**160.40**

Comparison: ABC substantially increases Product A's unit cost (€155.00 → €168.65) and Product C's (€127.00 → €160.40), while reducing Product B's (€147.00 → €126.77). Product A, despite requiring only 1 machine hour per unit, generates a full batch of production for every 200 units and consumes significant quality-inspection cost per unit produced; Product C, with a small batch size of 50, requires proportionately more set-ups (6 batches for only 300 units) and purchase orders than its modest machine-hour usage suggests. Traditional machine-hour-based costing captures neither of these batch- and order-driven cost patterns, undercosting both A and C at the expense of B.

Mark Allocation

Marks	Criteria
26–32	Traditional costing correctly calculated; number of batches correctly derived for each product; all four ABC cost pool rates correctly calculated; overhead correctly allocated to each product; ABC total cost and cost per unit correct for all three products.
19–25	Substantially correct with minor computational errors in one or two pool rates, batch calculations, or allocations.
10–18	Traditional costing correct; ABC approach understood but the batch-derivation step or rates/allocations contain multiple errors.
0–9	Limited understanding of either costing method; minimal correct workings.

Question 1(B)

8 marks

Consider the following view expressed at a recent management meeting of Fellows Ltd: "All organisations must take a long-term view if they are to survive and thrive therefore the relevant range does not matter in organisational decision making." State whether you agree or disagree, explaining the reason(s) for your answer.

Full Model Answer

DISAGREE

The conclusion does not follow from the premise. Taking a long-term strategic view does not eliminate the need to understand the relevant range — if anything, the relevant range remains central to sound decision-making at every time horizon, including long-term strategic decisions.

What the relevant range actually is, and why it still matters

The relevant range is the band of activity within which the assumed relationships between cost and volume (e.g. constant total fixed costs, constant variable cost per unit) are expected to hold. Every budget, cost estimate, and pricing decision — whether prepared for next month or as part of a five-year strategic plan — is only valid within some relevant range. Ignoring the relevant range does not become safer simply because the organisation is thinking long-term; it becomes *more* important, because long-term plans typically involve larger swings in activity level, making it more likely that assumptions valid at current volumes will break down (e.g. needing a second machine or additional supervisory staff, as with the step costs and expanded production discussed elsewhere in this paper).

Long-term strategic decisions are often precisely about moving between relevant ranges

Major long-term decisions — capacity expansion, entering a new product line (as Quaver Ltd considers with its guitar proposal in Question 3), or investing in new machinery — are frequently decisions about whether and how to move the organisation from its current relevant range into a new one, at a different, higher fixed cost base. Recognising the boundaries of the current relevant range, and planning for the step change in costs required to operate in a new one, is a core part of sound long-term strategic planning, not something that can be dispensed with because the organisation is 'thinking long-term.'

Conclusion: The relevant range concept and a long-term strategic outlook are not in conflict — the relevant range is a tool used within (and across) planning horizons to ensure cost and revenue assumptions remain valid, and understanding its limits is arguably most important precisely when an organisation is planning significant, long-term change.

Mark Allocation

Marks	Criteria
7–8	Clear disagreement with well-reasoned explanation of why the relevant range remains relevant regardless of time horizon, including recognition that long-term decisions often involve moving between relevant ranges.
5–6	Correct disagreement with reasonable explanation; one of the two supporting arguments underdeveloped.
2–4	Some relevant discussion but generic, or the link between relevant range and long-term decision-making not clearly made.
0–1	Agrees with the statement without meaningful justification, or no meaningful discussion provided.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2

Contessa Ltd

Total: 30 marks

Contessa Ltd commenced operations on 1 July 2020, manufacturing and selling a single product. Candidates are required to (A) prepare income statements under both marginal and absorption costing, (B) reconcile the resulting profit figures, and (C) distinguish planning activities from control activities.

Part	Requirement	Marks
(A)	Unit product cost, closing stock value and operating profit – marginal and absorption costing	18
(B)	Profit reconciliation note	6
(C)	Distinguish planning activities and control activities	6

Question 2(A) **18 marks**

Contessa Ltd, year to 30 June 2021: Sales 4,000 units; Selling price €24.60; Production 5,500 units; Variable manufacturing costs (total) €67,650; Fixed manufacturing overheads (total) €26,950; Variable selling expenses (total) €12,000; Fixed selling expenses (total) €8,000. (A) Calculate the unit product cost, the value of closing stock and the operating profit using (i) Marginal costing and (ii) Absorption costing.

Full Model Answer

Preliminary workings

Closing stock = Production − Sales = 5,500 − 4,000 = 1,500 units. Variable manufacturing cost per unit = €67,650 ÷ 5,500 = €12.30. Fixed manufacturing overhead per unit (absorption costing only) = €26,950 ÷ 5,500 = €4.90.

(i) Marginal costing

Unit product cost and closing stock – marginal costing

Item	€
Unit product cost (variable manufacturing cost only)	12.30
Closing stock value (1,500 units × €12.30)	**18,450**

Income statement – marginal costing

Item	€
Sales revenue (4,000 units × €24.60)	98,400
Less: Variable cost of goods sold (4,000 units × €12.30)	(49,200)
Less: Variable selling expenses	(12,000)
Contribution margin	**37,200**
Less: Fixed manufacturing overheads	(26,950)
Less: Fixed selling expenses	(8,000)
Operating profit (marginal costing)	**2,250**

(ii) Absorption costing

Unit product cost and closing stock – absorption costing

Item	€
Unit product cost (€12.30 variable + €4.90 fixed)	17.20
Closing stock value (1,500 units × €17.20)	**25,800**

Income statement – absorption costing

Item	€
Sales revenue (4,000 units × €24.60)	98,400
Less: Cost of goods sold (4,000 units × €17.20)	(68,800)
Gross profit	**29,600**
Less: Variable selling expenses	(12,000)
Less: Fixed selling expenses	(8,000)
Operating profit (absorption costing)	**9,600**

Reconciliation preview: The €7,350 profit difference (€9,600 − €2,250) equals the fixed manufacturing overhead carried forward in closing stock under absorption costing (1,500 units × €4.90 = €7,350), consistent with production (5,500 units) substantially exceeding sales (4,000 units) in this, the company's first year of trading.

Mark Allocation

Marks	Criteria
16–18	Both unit product costs, both closing stock values, and both fully-formatted income statements correct, with contribution margin and gross profit subtotals clearly shown.
12–15	Substantially correct with a minor computational error in one figure, or one method fully correct and the other largely correct.
7–11	Correct approach to at least one costing method; some errors or omissions in workings or presentation.
0–6	Limited understanding of the distinction between the two costing methods; minimal correct workings.

Question 2(B)

6 marks

Management at Contessa Ltd is confused as to why the profits reported differ under each of the methods in Part (A) above. Prepare a short note reconciling the profit figures and explaining why this difference occurs.

Full Model Answer

Why the profit figures differ

Absorption costing carries fixed manufacturing overhead forward in closing stock as part of the unit product cost, whereas marginal costing treats fixed manufacturing overhead as a period cost, charged in full against the current year regardless of how many units remain unsold. Because production (5,500 units) exceeded sales (4,000 units) in the year, more fixed overhead was deferred into closing stock under absorption costing than was charged to this year's profit, resulting in a higher reported profit under absorption costing.

Profit reconciliation

Item	€
Operating profit – marginal costing	2,250
Add: Fixed manufacturing overhead carried forward in closing stock (1,500 units × €4.90)	7,350
Less: Fixed manufacturing overhead carried forward in opening stock (none – first year of trading)	0
Operating profit – absorption costing	**9,600**

General rule: When production exceeds sales, as here, absorption costing reports a higher profit than marginal costing. When sales exceed production, marginal costing reports the higher profit. When production equals sales, the two methods report identical profit, since there is no change in stock to defer or release fixed overhead.

Mark Allocation

Marks	Criteria
5–6	Clear explanation of the fixed overhead deferral in stock under absorption costing, why production exceeding sales causes profit to differ, and a correct, clearly presented profit reconciliation.
3–4	Good explanation with a correct or substantially correct reconciliation; some lack of clarity or a minor computational error.
1–2	Some understanding shown of why the figures differ, but explanation is generic or the reconciliation is materially incomplete/incorrect.
0	No meaningful explanation or reconciliation provided.

Question 2(C) **6 marks**

Distinguish between planning activities and control activities in management accounting.

Full Model Answer

Planning

Planning is the process of establishing objectives and determining the means by which those objectives will be achieved; it is forward-looking and includes activities such as budgeting, forecasting, and setting targets for future periods (e.g. deciding on expected sales volumes, prices, and cost levels ahead of a new financial year).

Control

Control is the process of monitoring actual performance, comparing it against the plan (e.g. actual results against budget), identifying variances, and taking corrective action where necessary. It is a feedback process that follows on from planning — for example, comparing Contessa Ltd's actual production and sales volumes, and actual costs, against the figures budgeted at the start of the year, to identify and investigate any significant differences.

The planning-control cycle: Planning and control together form a continuous cycle: control information from one period (actual vs budget variances) feeds back into the planning process for the next period, helping management refine future budgets and targets based on what was learned.

Mark Allocation

Marks	Criteria
5–6	Both planning and control clearly and accurately distinguished, each with a correct definition and a relevant illustrative example.
3–4	Both concepts distinguished accurately with reasonable explanation; examples missing or underdeveloped.
1–2	One concept explained well; the other explained only partially or generically.
0	No meaningful distinction drawn.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3

Quaver Ltd

Total: 30 marks

Quaver Ltd is a small violin maker in Cork with demand often outstripping supply. Candidates are required to (A) calculate and explain core CVP measures for the 'Master' violin, (B) apply incremental analysis to a proposal to hire an apprentice, and (C) apply multi-product WACM breakeven analysis to a proposed new guitar product line, with a non-financial consideration.

Part	Requirement	Marks
(A)	Breakeven (units/revenue), operating profit, margin of safety, operating leverage – with explanation of use and CVP assumptions	15
(B)	Incremental analysis of the apprentice hire proposal	5
(C)	WACM breakeven and profit for the proposed guitar ('Lyric') line, with advice and a non-financial consideration	10

Question 3(A)

15 marks

Quaver Ltd, budgeted 'Master' violin for December 2021: Sales price €400/unit; Variable cost €220/unit; Budgeted fixed costs €9,000/month. (A) Calculate and explain the use of: (i) Breakeven point in units and in sales revenue; (ii) Operating profit, percentage margin of safety and degree of operating leverage if 65 Master units are actually sold. State two assumptions underpinning your analysis.

Full Model Answer

Contribution margin per unit = €400 − €220 = €180. Contribution margin ratio = €180 ÷ €400 = 45%.

(i) Breakeven point – 50 units, €20,000

Breakeven units = Fixed costs ÷ Contribution margin per unit = €9,000 ÷ €180 = **50 units**. Breakeven revenue = 50 × €400 = **€20,000** (or €9,000 ÷ 45% CM ratio). *Usefulness:* shows Quaver Ltd the minimum number of violins that must be sold each month before the business breaks even, useful for setting minimum monthly targets and assessing risk, particularly relevant given that Quaver Ltd's capacity (not just demand) may constrain how many units can realistically be produced.

(ii) Operating profit, margin of safety and operating leverage at 65 units sold

Operating profit = (65 × €180) − €9,000 = €11,700 − €9,000 = **€2,700**. *Usefulness:* shows expected profitability at the budgeted sales volume.

Working

Item	Value
Margin of safety % ((65 − 50) ÷ 65)	**23.08%**
Degree of operating leverage (Total CM €11,700 ÷ Operating profit €2,700)	**4.33**

Usefulness of margin of safety: shows how far sales can fall (from 65 units) before Quaver Ltd starts making a loss, giving a measure of the cushion in the December sales forecast. *Usefulness of operating leverage:* a leverage of 4.33 indicates that a 1% change in sales revenue would change operating profit by approximately 4.33%, highlighting that profit is highly sensitive to changes in sales volume at this level of activity.

Two assumptions underpinning the analysis

1. Costs can be reliably classified as fixed or variable and behave in a linear fashion within the relevant range — the €220 variable cost per violin and €9,000 fixed costs are assumed constant regardless of the exact number of units produced within a reasonable range around 50–65 units. **2.** All units produced are sold in the same period (no closing stock build-up), and the selling price of €400 is assumed constant regardless of the number of units sold (no volume discounts) — both reasonable short-term assumptions, though Quaver Ltd's supply-constrained market position (demand often outstripping supply) makes the 'units produced = units sold' assumption particularly likely to hold in practice.

Mark Allocation

Marks	Criteria
13–15	Breakeven, operating profit, margin of safety and operating leverage all correctly calculated with clear, specific explanations of usefulness; two relevant CVP assumptions correctly identified and explained.
9–12	Most figures correctly calculated with reasonable explanations; assumptions identified but underdeveloped.
5–8	Some figures correctly calculated; explanations generic; assumptions missing or only one given.
0–4	Limited understanding of CVP metrics; minimal correct workings.

Question 3(B) **5 marks**

Quaver Ltd is considering hiring an apprentice for the Cork workshop, at a cost of €18,000 per annum. This will enable Quaver Ltd to make and sell an additional 12 Master violins each month. Using the incremental approach, advise Quaver Ltd on whether they should proceed with this proposal.

Full Model Answer

Incremental analysis (monthly)

Item	€
Incremental revenue (12 units × €400)	4,800
Incremental variable cost (12 units × €220)	(2,640)
Incremental contribution margin	**2,160**
Incremental fixed cost (apprentice: €18,000 ÷ 12 months)	(1,500)
Incremental monthly profit	**660**

AGREE

Quaver Ltd should proceed with hiring the apprentice: the proposal generates a positive incremental profit of €660 per month (€7,920 per annum), since the additional contribution margin from the extra 12 units sold (€2,160 per month) exceeds the additional fixed cost of the apprentice (€1,500 per month).

Note: This positive result assumes Quaver Ltd can actually sell the additional 12 units each month — given that demand for violins is described as often outstripping supply, this appears to be a reasonable assumption, strengthening the case for proceeding.

Mark Allocation

Marks	Criteria
4–5	Correct incremental revenue, variable cost, and fixed cost calculated on a consistent (monthly or annual) basis; correct positive incremental profit figure; clear recommendation to proceed.
2–3	Substantially correct incremental analysis with a minor error, e.g. inconsistent monthly/annual basis; correct overall conclusion.
1	Some relevant figures identified but the incremental approach not correctly applied.
0	No meaningful incremental analysis or an incorrect recommendation.

Question 3(C) **10 marks**

Quaver Ltd is separately considering producing and selling guitars ('Lyric'): budgeted unit variable cost €180, budgeted unit sales price €350. Anticipated sales mix: 80% Masters, 20% Lyrics. Introducing Lyric would increase monthly fixed costs (total increase of €20,160 per annum). (i) Determine the breakeven point in units and in sales revenue. (ii) Determine the operating profit and advise Quaver Ltd on whether to implement the proposal if unit sales in December 2021 are 56 Masters and 14 Lyrics, with one non-financial consideration.

Full Model Answer – (i) WACM breakeven

Working – revised fixed costs and contribution per unit

Item	Value
Existing fixed costs (monthly)	€9,000
Additional fixed costs (€20,160 per annum ÷ 12)	€1,680
Total monthly fixed costs under the proposal	**€10,680**
Master contribution margin per unit	€180
Lyric contribution margin per unit (€350 − €180)	€170

Working – weighted average contribution margin (sales mix 80% Masters : 20% Lyrics, i.e. 4:1)

Item	Value
Contribution per 'basket' of 5 units (4 × €180 + 1 × €170)	€890
WACM per unit (€890 ÷ 5)	**€178.00**

Breakeven point – units and revenue

Item	Value
Total breakeven units (€10,680 ÷ €178)	60 units
Master breakeven (60 × 4/5)	48 units
Lyric breakeven (60 × 1/5)	12 units
Breakeven revenue (48 × €400 + 12 × €350)	**€23,400**

Full Model Answer – (ii) Operating profit and recommendation

Working – operating profit with the guitar proposal (56 Masters + 14 Lyrics)

Item	€
Master contribution (56 units × €180)	10,080
Lyric contribution (14 units × €170)	2,380
Total contribution margin	**12,460**
Less: Total monthly fixed costs under the proposal	(10,680)
Operating profit (with Lyric proposal)	**1,780**

Comparison – profit without the Lyric proposal (56 Masters only, original fixed costs)

Item	€
Master contribution (56 units × €180)	10,080
Less: Original monthly fixed costs	(9,000)
Operating profit (without Lyric proposal)	**1,080**

AGREE

Quaver Ltd should proceed with the Lyric guitar proposal: at the actual December sales mix of 56 Masters and 14 Lyrics, operating profit of €1,780 exceeds the €1,080 that would be earned from Master violin sales alone, an incremental improvement of €700 for the month.

> **Non-financial consideration**
> Quaver Ltd should weigh the opportunity cost of diverting limited workshop capacity and skilled labour away from the Master violin — a product for which demand already outstrips supply — toward a new, unproven guitar product. If producing Lyric guitars reduces the workshop's ability to fulfil existing (and already excess) demand for Master violins, this could damage customer relationships, extend violin waiting times, and harm Quaver Ltd's reputation in its core, already-successful product line, even though the financial analysis above appears favourable.

Mark Allocation

Marks	Criteria
8–10	WACM breakeven correctly calculated in units and revenue; operating profit at actual sales correctly calculated and compared against a sensible baseline; clear recommendation to proceed supported by the figures; a specific, well-reasoned non-financial consideration tied to Quaver Ltd's supply-constrained market position.
5–7	Substantially correct workings for (i) and (ii) with a minor error; reasonable recommendation and non-financial point given.
2–4	WACM approach understood but applied with errors; recommendation or non-financial consideration generic.
0–1	Limited understanding of the multi-product breakeven approach; minimal correct workings or recommendation.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4

Statement Evaluation

Total: 30 marks

Five independent statements covering conversion costs, sunk vs opportunity vs differential costs, operating leverage vs margin of safety in different cost structures, the high-low method, and cost structure vs sales mix. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Conversion costs – scope and definition	6
(B)	Sunk costs vs opportunity costs vs differential costs	6
(C)	Operating leverage and margin of safety: capital vs labour intensive	6
(D)	Alpine Ltd – high-low method cost estimate	6
(E)	Cost structure vs sales mix	6

Question 4(A)

6 marks

Conversion costs are all the costs involved in the production and sale of a unit of finished goods; they include both product costs and period costs.

Full Model Answer

DISAGREE

Conversion cost is defined much more narrowly than described: it is direct labour plus manufacturing overhead only.

What conversion cost actually covers

Conversion cost represents the cost of converting raw materials into a finished product: direct labour + manufacturing overhead. It deliberately **excludes** direct materials (which forms part of prime cost instead) and excludes selling and administrative costs entirely.

Why the statement is incorrect

The statement is wrong in two respects. First, conversion cost relates only to *production*, not to production **and sale** — selling costs (such as delivery, marketing, or sales commissions) are never part of conversion cost. Second, conversion cost consists entirely of product (manufacturing) costs; it does not include any period costs at all — period costs (such as selling and administrative expenses) are, by definition, expensed as incurred and are never part of the cost of a unit of product, whether measured as prime cost, conversion cost, or full production cost.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with an accurate definition of conversion cost (direct labour + manufacturing overhead only), correctly identifying both errors in the statement (inclusion of 'sale' and inclusion of period costs).
3–4	Correct disagreement with an accurate definition of conversion cost; only one of the two errors explicitly identified.
1–2	Correct disagreement but with a vague or partially inaccurate definition.
0	Incorrect verdict or no meaningful explanation.

Question 4(B)

6 marks

Sunk costs measure the benefit given up when one decision alternative is selected over another; thus, they are differential costs and should always be considered when making decisions.

Full Model Answer

DISAGREE

The statement is incorrect on two separate counts: the description given is actually the definition of an opportunity cost, not a sunk cost, and sunk costs are not differential costs — they are, in fact, the opposite: irrelevant to decision-making.

The description matches opportunity cost, not sunk cost

The potential benefit given up when one alternative is chosen over another is the definition of an **opportunity cost**. A sunk cost, by contrast, is a cost that has already been incurred in the past and cannot be recovered or changed by any current or future decision.

Sunk costs are not differential costs

A differential cost is one that differs between the alternatives being considered, making it relevant to the decision. A true sunk cost is, by definition, the **same** regardless of which alternative is chosen going forward (it has already happened and cannot be altered), which is exactly why sunk costs are irrelevant to decision-making and should be **ignored**, not 'always considered' as the statement claims.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement, correctly identifying that the statement's description matches opportunity cost rather than sunk cost, and correctly explaining that sunk costs are irrelevant (not differential) to decision-making.
3–4	Correct disagreement addressing one of the two errors clearly (either the opportunity cost mix-up or the differential cost claim).
1–2	Correct disagreement but with a vague or partially inaccurate explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(C)

6 marks

Operating leverage and margin of safety will be higher in a capital-intensive company than in a labour-intensive company with the same total revenues and total costs.

Full Model Answer

DISAGREE

Operating leverage will indeed tend to be higher in a capital-intensive company, but margin of safety will tend to be **lower**, not higher — the statement is only half correct.

Operating leverage

A capital-intensive cost structure (high fixed costs, low variable costs per unit) produces a higher contribution margin relative to operating profit at any given sales level than a labour-intensive cost structure (low fixed costs, high variable costs), because more of each sales euro is retained as contribution before fixed costs are covered. Operating leverage (Total CM ÷ Operating profit) is therefore **higher** in the capital-intensive company — this part of the statement is correct.

Margin of safety

A higher fixed cost base in the capital-intensive company increases the breakeven point (more contribution is required to cover the larger fixed cost base). Since margin of safety = (Actual sales − Breakeven sales) ÷ Actual sales, a higher breakeven point reduces the margin of safety. Margin of safety is therefore **lower**, not higher, in the capital-intensive company — contradicting the statement.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement identifying that operating leverage is indeed higher but margin of safety is lower (not higher) in a capital-intensive company, with correct reasoning linking fixed cost proportion to both measures.
3–4	Correct reasoning on one of the two measures with the other measure's direction incorrect or unexplained.
1–2	Some relevant understanding of cost structure and risk shown, but reasoning is generic or the verdict is unclear.
0	Incorrect verdict (agrees with the statement) or no meaningful engagement with the statement.

Question 4(D)

6 marks

Monthly activity levels and associated costs at Alpine Ltd: HIGH: June, 3,450 units, total cost €14,240. LOW: November, 2,100 units, total cost €9,920. If it is planned to produce 2,500 units in December 2021, then based on the cost function for the most recent financial period, the expected cost is €11,200.

Full Model Answer

AGREE

The expected cost of €11,200 given in the statement is correct.

Working – high-low method

Item	Value
Variable cost per unit ((€14,240 − €9,920) ÷ (3,450 − 2,100))	(€4,320 ÷ 1,350) = €3.20 per unit
Fixed cost (at high point: €14,240 − (3,450 × €3.20))	€3,200
Fixed cost (at low point: €9,920 − (2,100 × €3.20)) – check	€3,200
Cost function	Total cost = €3,200 + (€3.20 × units)

Estimated cost at 2,500 units (December 2021)

Item	Value
€3,200 + (€3.20 × 2,500)	€3,200 + €8,000
Estimated cost	**€11,200**

Mark Allocation

Marks	Criteria
5–6	Correct agreement; variable cost per unit and fixed cost correctly calculated and cross-checked at both points; correct cost function applied to confirm €11,200.
3–4	Correct agreement with a minor computational error, or the cross-check at the second point omitted.
1–2	High-low method attempted with the correct general approach but a significant computational error.
0	Incorrect verdict (disagrees with the statement) or no meaningful workings.

Question 4(E)

6 marks

Cost structure refers to the relative proportions in which an organisation's products are sold.

Full Model Answer

DISAGREE

The description given in the statement is the definition of **sales mix**, not cost structure.

What cost structure actually is

Cost structure refers to the relative proportion of fixed costs and variable costs within an organisation's total costs. As explored in Part (C) of this question, a cost structure weighted towards fixed costs (capital intensive) produces higher operating leverage and a lower margin of safety than one weighted towards variable costs (labour intensive), for the same total revenue and total costs.

What the statement is actually describing

The relative proportions in which an organisation's different products are sold is the definition of **sales mix**. Sales mix is central to multi-product breakeven analysis using the weighted average contribution margin approach, as applied to Quaver Ltd's Master and Lyric proposal in Question 3. Cost structure and sales mix are both important CVP concepts, but they describe entirely different things: one relates to the fixed/variable split of costs, the other to the mix of products sold.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with an accurate definition of cost structure and a correct identification that the statement's description actually matches sales mix.
3–4	Correct disagreement with a reasonable definition of cost structure; the sales mix point not explicitly identified.
1–2	Correct disagreement but with an inaccurate or vague definition of cost structure.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS